

PA2 Report - Requirement 2: User Analysis

Group: 06

Course: CSC13112 - UI/UX Design

Assignment: Project Assignment 2

Product Scope: Freestyle Chess mobile website on smartphone browser

Document Purpose: User problem identification, idea generation, affinity diagramming, and problem prioritization

1. Overview & Brainstorming Setup

1.1 Objectives

The goal of this phase is to analyze the empirical evidence collected from Requirement 1 (72 survey responses and 13 direct interview sessions), brainstorm to identify core user problems, generate potential solution ideas, organize them using an **Affinity Diagram**, and prioritize the most severe problems to solve in the subsequent design proposal.

1.2 Brainstorming Session Record

Field	Details
Date & Time	July 23, 2026
Participants	Group 06 Members (Lê Mai Hoài Bảo, Trương Công Thiên Phú, Lâm Hữu Khánh, Phạm Chí Bảo Ninh, Phùng Ngọc Tuấn)
Tool Used	Digital Sticky Notes via Figma / FigJam
Input Evidence	06-PA2-UserResearch.md, Khảo sát trải nghiệm người dùng trên Freestyle Chess mobile web (Responses) - Form Responses 1.csv, Interview clip recordings
Deliverable Output	Affinity Diagram & Prioritized Problem Matrix

2. Affinity Diagram

2.1 Visual Affinity Board (Digital Sticky Notes)

Below is the structured Affinity Board generated during the group brainstorming session, grouping user observations, pain points, and potential solution ideas into logical clusters.

Ergonomic Navigation & One-Handed Reachability

Hamburger menu
top-left hard to
reach one-
handed

Users stretch
thumb / shift grip
to tap nav

Persistent bottom
navigation bar

FAB / bottom-
right drawer for
reachability

Homepage Onboarding & Value Proposition

Hero section
cluttered, no
onboarding

New users don't
understand site
purpose

Minimalist hero
section + clear
CTA

"What is
Freestyle Chess?"
quick modal
guide

Search, Filtering & Information Retrieval

Rating list: no
sticky search,
repetitive
scrolling

Can't search
news/events by
keyword

Schedule lacks
match details

Sticky search bar
fixed at top

Filter chips +
auto-suggest
search

Surface richer
schedule context
in item cards

Video Player Interaction & Feedback

No loading
feedback when
tapping video

New video
doesn't pause old
video (audio
overlap)

Single-active-
player logic
(auto-pause old
video)

Skeleton loaders
on video tap

Link to interactive FigJam board: [FigJam Affinity Diagram Canvas](#)

2.2 Raw Sticky Notes Inventory

Notes: Data in the "Source / Evidence" column supports the "Stick Note Content", exceptions are listed below the table.

Note ID	Category	Source / Evidence	Sticky Note Content
N01	Problem	Survey Q12, P01/P02/P03/P04/P05/P07/P08	Mobile Hamburger menu is placed at top-left, making one-handed thumb interaction painful/annoying.
N02	Problem	Survey Q13 (43.1%), P02/P03/P04/P05/P07/P08	Users must stretch fingers, shift phone grip, shift the phone, or use two hands to tap navigation items.
N03	Idea	Brainstorm	Move primary navigation to a persistent Bottom Navigation Bar.
N04	Idea	Brainstorm, P05	Provide a floating action button (FAB) or bottom-right drawer for one-handed reachability.

Note ID	Category	Source / Evidence	Sticky Note Content
N05	Problem	Survey Q09 (avg 3.68/5), P03/P05/P08	Homepage Hero section is cluttered with banners and lacks onboarding explaining Freestyle Chess format to new users.
N06	Problem	Survey Q10 (51.4%), P03/P05/P08	New users feel disoriented and do not understand website purpose upon landing.
N07	Idea	Brainstorm	Redesign Hero section with a minimalist intro banner, clear onboarding text, and prominent CTA.
N08	Idea	Brainstorm	Add an interactive "What is Freestyle Chess?" quick modal guide for newcomers.
N09	Problem	Survey Q17 (81.9%), P02/P05/P06/P08	Rating leaderboard is an infinite scroll list with no sticky search bar, causing repetitive scrolling.
N10	Problem	Survey Q19 (61.1%), P02/P04/P06/P07	Users cannot search news/events by keywords and must browse long chronological lists.
N11	Idea	Brainstorm	Add a Sticky Search Bar fixed at top of long listing pages (Rating, News, Videos).
N12	Idea	Brainstorm	Provide filter chips (by major event, date, rating range) and auto-suggest search queries.
N13	Problem	Survey Q21 (52.8%), P02/P03/P05/P06/P07/P09	Video player lacks immediate visual feedback (loading spinner) when tapped, leading to rage clicks.
N14	Problem	Interview P09/P10 clips; P04/P05/P07	Playing a new video does not pause the previous video, resulting in overlapping audio channels.
N15	Idea	Brainstorm; P04/P05/P07/P09	Enforce a Single-Player audio/video model (playing new video automatically stops old video).
N16	Idea	Brainstorm; P02/P03/P09	Add skeleton loaders and loading spinners instantly upon tapping video play controls.
N30	Problem	P09 (see also N20/N21/N24/N28/N29)	Schedule only shows date and location; dedicated followers cannot find match time, participants, format, or bracket info, and searched elsewhere (including an external site) before giving up.
N31	Idea	Brainstorm; P09	Surface richer match context (time, players, format, bracket) directly within Schedule items

Note ID	Category	Source / Evidence	Sticky Note Content
			via an expandable popup or dedicated calendar view.

Exceptions:

- P08 Neutral about N10.
- P08 supports N15 but neutral about N14.
- P06 doesn't think N14 is a problem, and actually like it that way.

Additional Problem Outside the Survey:

Notes: These are problems that arose from direct Interviews rather than from questionnaires, worth considering.

Note ID	Category	Source / Evidence	Sticky Note Content
N17	Problem	P01	Confirmation dialogs are too small, forcing users to scroll inside the dialog to read the full message before making a decision.
N18	Problem	P01	Page transitions provide no loading indicator, making users unsure whether the site is loading or frozen and encouraging repeated taps.
N19	Problem	P02	Video & Streams carousel looks swipeable because thumbnails are cut off, but horizontal swiping does not work and users must use Prev/Next buttons instead.
N20	Problem	P02	Schedule navigation uses a down-arrow icon suggesting expansion, but the content slides in from the side, causing confusion.
N21	Problem	P02	Schedule has two different exit controls ("X" and "Back to Main Menu"), making it unclear which one to use.
N22	Problem	P05	Rating list rows have overly similar typography/visual structure, making it easy to overlook the desired player while scanning.
N23	Problem	P06/P09	Rating search requires the player's name in a specific format/order, causing searches to fail when users enter a natural first-name/last-name order.
N24	Problem	P09	Schedule does not provide enough information about upcoming matches or tournament brackets, forcing users to look elsewhere for this information.
N25	Problem	P09	Video carousel stops responding to swipe gestures once a video is playing, forcing users to use Prev/Next controls instead.
N26	Problem	P10	Hero video "Stop" button does not actually stop the video OR restart the video from the beginning instead of correctly pausing/stopping it. Button state does not match the video's actual state

Note ID	Category	Source / Evidence	Sticky Note Content
N27	Problem	P10	Ranking search cannot search by chess title or rating, despite users naturally wanting to find players using those attributes.
N28	Problem	P10	Schedule banners appear interactive/tappable but do nothing when tapped.
N29	Problem	P10	Schedule menu has a broken scroll boundary that prevents users from fully viewing some of its content.

2.3 Affinity Cluster Descriptions

Cluster A: Ergonomic Navigation & One-Handed Reachability

- **Core Problem:** The current hamburger menu icon is positioned at the top-left corner. On modern smartphone screens, this forces right-handed users (44.4% one-handed users) to stretch their thumb unnaturally or change their grip, increasing device drop risk.
- **Generated Ideas:**
 - Implement a mobile **Bottom Navigation Bar** with key tabs (Home, News, Schedule, Rating, Videos).
 - Place a floating drawer or menu button at the bottom-right corner.

Cluster B: Homepage Onboarding & Value Proposition Clarity

- **Core Problem:** 52.8% of surveyed users had never heard of Freestyle Chess. The current homepage Hero section is overloaded with event banners without explaining the unique rules/format of Freestyle Chess, causing cognitive fatigue.
- **Generated Ideas:**
 - Simplify the Hero section hierarchy.
 - Add a prominent onboarding banner with a concise explanation ("Fischer Random / Chess 960 format") and a primary CTA ("Explore Events").

Cluster C: Search, Filtering & Information Retrieval

- **Core Problem:** 61.1% of users reported difficulty finding specific info when they don't remember exact names. Long rating lists require 20+ seconds of vertical scrolling without a sticky search bar or pagination. This same theme extends beyond the Rating/News lists to the Schedule section: a dedicated follower (P09) actively searched the site for match time, participants, and bracket context that simply isn't surfaced anywhere, and gave up only after also checking an external source (N30).
- **Generated Ideas:**
 - Add a **Sticky Search Bar** that remains visible while scrolling.
 - Implement **Quick Filter Chips** (e.g., "Geller Cup", "Grandmaster Rating") and Auto-suggest queries.
 - Surface richer match context (time, players, format, bracket) directly within Schedule items so dedicated followers don't need to search elsewhere (N31).

Cluster D: Video Player Interactions & Feedback

- **Core Problem:** Tapping video thumbnails provides no loading indicator, causing users to think the site is frozen. Multiple videos can play simultaneously, causing overlapping audio streams.
 - **Generated Ideas:**
 - Enforce **Single-Active-Player logic** (auto-pause active video when another starts).
 - Integrate skeleton loaders and instant visual press states.
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3. Problem Prioritization Matrix

To scientifically select the most critical problems for the design proposal phase, each identified problem was evaluated on a scale from 1 to 5 using the following strict rubric:

3.1 Evaluation Rubric

- **Frequency:**
 - **1:** < 10% users experienced it.
 - **2:** 10% - 30% users experienced it.
 - **3:** 30% - 50% users experienced it.
 - **4:** 50% - 70% users experienced it (Very common).
 - **5:** > 70% users experienced it (Almost everyone).
- **Severity (Based on Nielsen's Severity):**
 - **1:** Cosmetic issue (Minor visual flaw, no real impact on task).
 - **2:** Minor usability problem (Slight annoyance but easily overcome).
 - **3:** Moderate problem (Causes hesitation or extra steps, but task is completed).
 - **4:** Major problem (Severe disruption, user gets frustrated or task takes significantly longer).
 - **5:** Usability catastrophe (Task blocked, user gives up or clicks away).
- **Evidence Strength:**
 - **1:** Assumption or gut feeling (No data).
 - **2:** Mentioned by only 1 interview participant, no survey data.
 - **3:** Backed by minor survey data OR 2-3 interview clips.
 - **4:** Backed by strong survey data AND clear interview observations.
 - **5:** Overwhelming proof (High survey % AND repeatedly observed failing in video clips).
- **Feasibility:**
 - **1:** Requires total system architecture rewrite (Very Hard).
 - **2:** Requires major backend/API logic changes.
 - **3:** Requires creating entirely new complex UI screens.
 - **4:** Requires moderate frontend adjustments (e.g., adding sticky CSS, new filters).
 - **5:** Quick fix (e.g., repositioning a button, changing text, fixing CSS limits - Very Easy).

Priority Score = Frequency + Severity + Evidence Strength + Feasibility

Problem ID	Problem Description	Frequency (1-5)	Severity (1-5)	Evidence Strength (1-5)	Feasibility (1-5)	Priority Score (4-20)	Decision
P-01	One-handed reachability issues with top-left hamburger menu	4	5	5	5	19	Selected (High)
P-02	Lack of sticky search & filtering on long listing pages (Rating & News)	5	4	5	4	18	Selected (High)
P-03	Overlapping video audio streams & lack of loading feedback	4	4	5	5	18	Not proposal focus
P-04	Overloaded Hero section & lack of onboarding for first-time users	4	3	4	5	16	Lower priority
P-05	Schedule lacks sufficient match/tournament information for dedicated followers	1	4	2	3	10	Noted (Below Threshold)

3.1 Scoring Rationale & Justification

1. P-01 (One-handed navigation reachability - Score 19/20):

- *Frequency (4/5)*: 44.4% of surveyed users operate smartphones with one hand, and 43.1% reported physical effort/reachability issues.
- *Severity (5/5)*: Crucial navigation failure; top-left menu placement forces grip shifts and thumb stretching, creating high device drop risks during mobile commuting.
- *Evidence Strength (5/5)*: Validated by survey Q12/Q13 and direct video clip observations (P01, P02, P03).
- *Feasibility (5/5)*: Highly feasible to solve by introducing a persistent mobile Bottom Navigation Bar or floating action drawer.

2. P-02 (Lack of sticky search & filtering on long lists - Score 18/20):

- *Frequency (5/5)*: 61.1% of users reported search difficulty when exact names are unknown; 81.9% requested search/filtering for long lists.

- *Severity (4/5)*: Forces 20+ seconds of repetitive vertical scrolling, severely degrading short 2-3 minute browsing sessions.
- *Evidence Strength (5/5)*: Validated by survey Q16/Q19/Q17 and interview clip observations (P01, P02, P04).
- *Feasibility (4/5)*: Feasible via sticky position CSS, query filter chips, and auto-suggest input handling.

3. P-03 (Video player feedback delay & audio overlap - Score 18/20):

- *Frequency (4/5)*: 52.8% of users expected old videos to auto-stop; 44.4% were confused by unresponsive video tap states.
- *Severity (4/5)*: Overlapping audio channels create extreme cognitive discomfort; missing loading spinners trigger repeated rage clicks.
- *Evidence Strength (5/5)*: Validated by survey Q20/Q21/Q22 and interview clip observations (P02, P03, P09).
- *Feasibility (5/5)*: Highly feasible via single-player event logic and skeleton loader components.

4. P-04 (Overloaded Hero section & onboarding gap - Score 16/20):

- *Frequency (4/5)*: 52.8% of surveyed participants were first-time users who had never heard of Freestyle Chess.
- *Severity (3/5)*: Disorients newcomers upon landing, though experienced users can scroll past it.
- *Evidence Strength (4/5)*: Validated by survey Q09/Q10 and interview clip observation (P03 L.T.K).
- *Feasibility (5/5)*: Highly feasible by reorganizing content hierarchy and adding a concise intro banner.

5. P-05 (Schedule lacks sufficient match/tournament information - Score 10/20):

- *Frequency (1/5)*: Reported by only one participant (P09) across all 82 research participants; no survey question directly targeted Schedule information completeness, so this cannot be scored higher on breadth of evidence alone.
- *Severity (4/5)*: P09 actively searched the Schedule page, the tournament's own info page, and even followed an external link before giving up entirely — indicating a real, unmet need rather than a passing inconvenience (N30, N20, N21, N24, N28, N29).
- *Evidence Strength (2/5)*: By the rubric's letter, this is "mentioned by only 1 interview participant, no survey data." However, the group weights this qualitatively: P09 is a self-identified active chess follower who deliberately sought match brackets and player pairings — exactly the information-seeking behavior of a core, retained user — whereas most other survey/interview respondents are first-time or casual users still forming a basic impression of the site. A gap that blocks a dedicated follower is judged more consequential to long-term retention than its raw frequency score suggests, which is why the group chose to carry P-05 into the Proposal despite the low quantitative score.
- *Feasibility (3/5)*: Requires new UI screens (e.g., a calendar view) to meaningfully close the gap; a lighter popup-based fix is more feasible but does not fully resolve the underlying information gap.

4. Final Problem Statements

Problem Statement 1 (Navigation & Reachability)

Mobile users need the primary navigation to sit within natural thumb reach **because** the hamburger menu is anchored top-left, forcing users to stretch their thumb or shift their grip just to tap a nav item, **especially when** browsing one-handed while on the move.

Problem Statement 2 (Information Retrieval & Search)

Mobile users need a fixed, filterable way to search long content lists **because** the Rating list has no sticky search bar and News/events can't be searched by keyword, leaving users to scroll repetitively through long, unfiltered lists, **especially when** they don't remember the exact name or date they're looking for.

Problem Statement 3 (Media Interaction & Feedback)

Mobile users need instant tap feedback and single-video playback **because** tapping a video gives no loading indicator and starting a new video doesn't pause the one already playing, causing overlapping audio, **especially when** watching match highlights or livestreams back-to-back.

Problem Statement 4 (Schedule Information Depth)

Dedicated chess followers need enough match and tournament context directly within the Schedule section **because** the current view shows only date and location, forcing users to search the tournament's own page or leave the site entirely to find who is playing or how matches are structured, **especially when** they already know the format and are trying to plan around specific upcoming matches.

5. Conclusion & Transition to Project Proposal

The user analysis phase successfully consolidated the raw empirical data from Requirement 1 into four clear affinity clusters, prioritized the key problems using a structured score matrix, and formulated four final Problem Statements.

These problem statements will serve as the direct foundation for generating conceptual design solutions in **Requirement 3 (Project Proposal)**.